Muhammad Talha Zafar

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PROFILE

I am a first-year Mechanical Engineering student at NUST Islamabad with a strong academic foundation and a passion for precision design. Having recently completed my foundational semesters with a focus on problem-solving and technical drawing, I am eager to transition from theory to practice. I am driven by a deep interest in aerospace innovation and am seeking an internship where I can contribute to technological growth while immersing myself in real-world engineering projects alongside industry leaders. I am also open to opportunities in digital marketing.

EDUCATION

Bachelors in Mechanical Engineering 2025 – Present
National University of Science and Technology (NUST), Islamabad
Field of Study: Mechanical Engineering

- Currently pursuing a Bachelor's degree in Mechanical Engineering at NUST, one of Pakistan's top-ranked engineering universities, with a focus on precision design, problem-solving, and applied engineering.

Intermediate | Grade: A+ 2023 – 2025
Govt. Comprehensive Higher Secondary School, Sahiwal
Pre-Engineering

- Completed Intermediate building a strong foundation in Mathematics, Physics, and Chemistry for engineering studies.

Matriculation | Grade: A+ 2020 – 2022
Sanabil Educational Complex, Dad Fatyana, Chichawatni
Science

- Completed Matriculation (Science) with strong academic grounding in Physics, Chemistry, Mathematics, and Biology.

EXPERIENCE

Technical Product Marketing & Automation
Remote
Digital Marketing - Machine Learning - Software

- Successfully engineered a professional brand identity, MZ Robotics, and designed a comprehensive B2B digital marketing funnel for a specialized industrial robotics product. By leveraging HubSpot CRM, developed and managed an automated email nurturing campaign that prioritized technical communication to drive lead conversion. Through the rigorous analysis of engagement metrics, deployed data-driven content to effectively bridge the gap between complex engineering design and strategic customer acquisition.

- Conducted a zero-budget digital competitive analysis for the industrial robotics sector, benchmarking MZ Robotics against global market leaders. Utilized SimilarWeb to analyze competitor traffic sources and map organic search strategies, identifying 5 high-value keyword gaps for organic growth. Authored a Strategic Positioning Report that identified cost-effective SEO opportunities to differentiate from established competitors without paid media spend

PROJECTS

Automatic Parking System 2024
Arduino / Embedded Systems / C++

- Designed and prototyped an automated parking solution using Arduino Uno and IR sensors to detect vehicle occupancy. Integrated servo motors for automated gate control and an I2C LCD for real-time slot availability updates.

IoT-Based Automated Irrigation System 2024
Arduino / IoT / C++

- Engineered a self-regulating watering system centered on an Arduino Uno and soil moisture sensors (hygrometers). Programmed a system in C++ to trigger a 5V relay and DC water pump based on real-time soil data.

Rube Goldberg Engineering Challenge 2024
Mechanical Design / Prototyping

- Developed a Rube Goldberg machine utilizing principles of energy transfer and mechanical advantage. Designed a sequence of complex chain reactions to complete a simple task, demonstrating proficiency in physics, structural stability, and iterative prototyping.

Driver Drowsiness Alarm 2024
Embedded Systems / C++ / Wearable Tech

- Prototyped wearable glasses with IR sensors to detect eye closure and microsleep. Designed a C++ logic loop to trigger an audible alarm, enhancing road safety via real-time monitoring.

Autonomous Line-Following Robot (LFR) 2024
Robotics / Arduino / Embedded Systems

- Engineered a high-speed autonomous robot using an Arduino Uno and an IR sensor array for precision path tracking.

Autonomous Pipe-Climbing Robot 2025
Robotics / Mechanical Design / ESP32

- Engineered a vertical climbing robot utilizing an ESP32-based control system and high-torque motors. Developed a spring-loaded four-wheel mechanism to maintain constant normal force against the pipe surface.

SKILLS

AutoCAD | Fusion 360| SolidWorks |Proteus | Arduino IDE| ROS 2 | VS Code |PlatformIO | Dev C++| HubSpot CRM |SimilarWeb | Microsoft Office (Word, Excel, PowerPoint)

SOFTWARE & TOOLS

Website Development| Digital Marketing |Machine Learning | Engineering Drawing | 3D Modeling | Arduino Programming |Good Communication | B2B Marketing |SEO & Competitive Analysis

PROGRAMMING LANGUAGES

C / C++ | HTML / CSS| Java |Python

AWARDS

STEM Competition Winner **2025**
District Sahiwal